

Implementation of a Two-Electrode Fixed Potential-Difference Method in OpenMX 3.9.9

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1 Purpose and scope

This extension adds a minimal fixed electrode potential-difference capability to OpenMX 3.9.9. Two sets of atoms define left and right electrodes. The fixed input V_c applies $+V_c/2$ to the left group and $-V_c/2$ to the right group; the constraint-potential difference is therefore V_c . The transferred charge is determined self-consistently.

The implementation covers input parsing, electrode assignment, Mulliken population sums, an unconstrained reference state, the constraint Hamiltonian, fixed- V_c convergence tests, and output. It does not implement target-charge feedback, constraint forces, constraint-energy corrections, restart persistence, multiple constraints, or NEGF coupling.

2 Definitions and derivation

2.1 Mulliken populations and transferred charge

Let $P_{\mu\nu}$ and $S_{\mu\nu} = \langle \phi_\mu | \phi_\nu \rangle$ denote the density and overlap matrices in a non-orthogonal atomic-orbital basis. The Mulliken electron population assigned to atom A is

$$N_A = \sum_{\mu \in A} \sum_{\nu} P_{\mu\nu} S_{\nu\mu}. \quad (1)$$

The electrode populations are

$$N_L = \sum_{A \in L} N_A, \quad N_R = \sum_{A \in R} N_A. \quad (2)$$

An ordinary $V_c = 0$ calculation is converged first. Its populations N_L^0, N_R^0 define

$$\Delta N_L = N_L - N_L^0, \quad \Delta N_R = N_R - N_R^0. \quad (3)$$

Positive transfer is defined as electron loss from the left and gain by the right electrode:

$$Q = \frac{-\Delta N_L + \Delta N_R}{2}. \quad (4)$$

For a charge-conserving two-electrode system, $\Delta N_L + \Delta N_R = 0$ and Eq. (4) reduces to $Q = -\Delta N_L = \Delta N_R$. The factor $1/2$ prevents double counting.

2.2 Constrained functional

Introduce an electrode weight

$$w(\mathbf{r}) = \begin{cases} +1, & \mathbf{r} \in L, \\ -1, & \mathbf{r} \in R, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

A functional containing the fixed external electrode potential is

$$W[n; V_c] = E_{\text{DFT}}[n] + \frac{V_c}{2} \int w(\mathbf{r}) n(\mathbf{r}) d\mathbf{r}. \quad (6)$$

Variation with respect to n adds

$$V_{\text{con}}(\mathbf{r}) = \frac{V_c}{2} w(\mathbf{r}) \quad (7)$$

to the Kohn–Sham potential. V_c is fixed input rather than a varied Lagrange multiplier. The induced transfer is measured using the Mulliken atomic partition of Eq. (4).

A positive V_c raises electron levels on the left and lowers them on the right. Thus positive V_c drives the positive charge-transfer convention.

2.3 Euler-Lagrange equation and ensemble

The semicolon in $W[n; V_c]$ is important: V_c is prescribed and variation is performed only with respect to the density, subject to the ordinary total electron-number normalization. Introducing the chemical potential μ for that normalization gives

$$\frac{\delta}{\delta n(\mathbf{r})} \left\{ E_{\text{DFT}}[n] + \frac{V_c}{2} \int w n d\mathbf{r} - \mu \left(\int n d\mathbf{r} - N_e \right) \right\} = 0, \quad (8)$$

or

$$\frac{\delta E_{\text{DFT}}}{\delta n(\mathbf{r})} + \frac{V_c}{2} w(\mathbf{r}) = \mu. \quad (9)$$

The self-consistent density is therefore stationary for a fixed external electrode field. This is distinct from target-charge cDFT, where V_c is also varied and $\partial W / \partial V_c = 0$ enforces a specified population. No such second stationarity condition or target population exists here.

For conserved total charge, Eq. (4) implies

$$N_L - N_R = \text{constant} - 2Q. \quad (10)$$

The field-dependent part of the reduced functional is consequently

$$\mathcal{F}(Q; V_c) = E_{\text{DFT}}(Q) - V_c Q + \text{constant}. \quad (11)$$

Stationarity with respect to the induced transfer gives

$$\frac{dE_{\text{DFT}}}{dQ} = V_c. \quad (12)$$

Thus V_c is the field conjugate to Q and the imposed constraint-potential difference. It is not the self-consistent Hartree voltage $\Delta\phi_H$, which must be read independently from the Hartree grid.

Equation (9) establishes density stationarity for the implemented one-electron problem. A stronger statement about OpenMX total energies and forces would require adding $\text{Tr}[PH^{\text{con}}]$ to the reported energy with consistent double-counting treatment, and differentiating the overlap-dependent matrix with respect to atomic positions. Those energy, force, and stress terms are not implemented. Consequently fixed-geometry densities, charges, and potentials are supported, but constrained energy comparisons and geometry optimization are not variationally supported.

3 Constraint matrix in a non-orthogonal basis

Suppose the local constraint takes atom-centered values V_A . The exact two-center matrix element is

$$\langle \phi_{A\mu} | V_{\text{con}} | \phi_{B\nu} \rangle = \int \phi_{A\mu}^*(\mathbf{r}) V_{\text{con}}(\mathbf{r}) \phi_{B\nu}(\mathbf{r}) d\mathbf{r}. \quad (13)$$

Approximating the potential in the overlap region by the symmetric endpoint average yields

$$H_{A\mu,B\nu}^{\text{con}} \simeq \frac{V_A + V_B}{2} \int \phi_{A\mu}^*(\mathbf{r}) \phi_{B\nu}(\mathbf{r}) d\mathbf{r} \quad (14)$$

$$= \frac{V_A + V_B}{2} S_{A\mu,B\nu}. \quad (15)$$

This expression is Hermitian and reduces to a diagonal on-site shift in an orthogonal basis. The atom value is

$$V_A = \begin{cases} +V_c/2, & g_A = 1, \\ -V_c/2, & g_A = 2, \\ 0, & g_A = 0. \end{cases} \quad (16)$$

Input values are in eV and are divided by 27.2113845 before addition to the Hartree-unit OpenMX Hamiltonian.

The scalar potential is added to both spin-diagonal blocks in a non-collinear calculation and not to spin-off-diagonal blocks. It is added before orbital contraction so that the existing contraction path transforms the complete Hamiltonian consistently.

4 Fixed potential-difference control

The robust sweep mode is enabled by

```
1 Potential.Constraint.Vc 0.24
```

The code converges the $V_c = 0$ reference, switches on the specified fixed potential, waits at least three constrained iterations, and then terminates when the electronic-energy criterion is met. The finite polarization itself can leave a nonzero density metric, so that metric is not used as the sole fixed- V_c termination condition.

Independent calculations provide $Q(V_c)$ and the Hartree response. There is no target-charge, proportional-feedback, or variable- V_c input mode.

5 Hartree sign and capacitance

OpenMX dVHart is an electron potential energy entering the one-electron Hamiltonian. Since the electron charge is $-e$, the physical electrostatic potential is

$$\phi(\mathbf{r}) = -\frac{V_H(\mathbf{r})}{e}. \quad (17)$$

Thus

$$\Delta\phi_{R-L} = -\frac{V_H(R) - V_H(L)}{e}. \quad (18)$$

Correcting this sign reverses voltage polarity but does not alter the positive capacitance magnitude

$$C_{\text{total}} = \frac{|Q|e}{|\Delta\phi|}, \quad \frac{C_{\text{total}}}{A} = \frac{|Q|e}{|\Delta\phi|A}. \quad (19)$$

The analysis program plane-averages the Hartree cube,

$$\bar{V}_H(x) = \frac{1}{A} \int_A V_H(x, y, z) dy dz, \quad (20)$$

and interpolates it at the two electrode coordinates.

The fixed external constraint difference is not automatically identical to the Hartree voltage. If an electrochemical voltage is required, the contribution conjugate to charge transfer must be defined thermodynamically; identifying it with V_c , or combining it with the Hartree drop, requires an explicit and consistent sign convention.

6 Input syntax

```
1 Potential.ConstraintDFT      on
2 Potential.Constraint.Vc      0.24
3 Potential.Constraint.Output  on
4
5 <Potential.Constraint
6 1 1
7 2 1
8 3 2
9 4 2
10 Potential.Constraint>
```

Group 0 is unconstrained, group 1 is the left electrode, and group 2 is the right electrode. Omitted atoms remain in group 0. The parser validates atom indices, group values, and the existence of both electrode groups.

7 Detailed source modifications

7.1 openmx_common.h

The header defines enable/output flags, the atom-group array, reference-state flags and iteration, fixed V_c , current and reference populations, transferred charge, and the function prototypes.

7.2 Input_std.c

New scalar keywords are read with backward-compatible defaults. After the normal atom arrays are allocated, PotentialConstraintGroup is allocated and initialized to zero. The independent block is parsed after the standard coordinates. NEGF is rejected because it is outside the MVP.

7.3 Set_Hamiltonian.c

After construction of the ordinary uncontracted Hamiltonian, the code evaluates the atom values V_A, V_B and applies Eq. (15):

```
1 vA = group[A]==1 ? +Vc/2 : group[A]==2 ? -Vc/2 : 0.0;
2 vB = group[B]==1 ? +Vc/2 : group[B]==2 ? -Vc/2 : 0.0;
3 dv = 0.5*(vA + vB)/eV2Hartree;
4 H[spin][A][B][mu][nu] += dv*OLP[0][A][B][mu][nu];
```

The constraint is inactive until the reference state has converged.

7.4 Potential_Constraint.c

This new file contains population aggregation, initialization, reference capture, fixed- V_c reporting, and final reporting. OpenMX has already gathered per-atom Mulliken populations to the host after Mulliken_Charge(); the host sums the groups and broadcasts the two totals to all ranks.

7.5 DFT.c

The DFT driver initializes the state, evaluates the populations after every Mulliken analysis, and captures the first converged unconstrained reference. The maximum SCF count is doubled to accommodate reference and constrained stages. The fixed- V_c calculation uses its dedicated convergence rule and minimum post-reference iteration count. A final summary is emitted after the final Mulliken analysis.

7.6 makefile and analysis programs

Potential_Constraint.o is added to the OpenMX object list. The minimal distribution patch does not alter compiler or library paths. capacitance_from_vhart.py performs cube averaging, sign conversion, and unit conversion. plot_v_ct.py collects a sweep into CSV and creates Q-V and V- C_{total}/A figures.

8 Positive and negative sweep for $d = 10.05 \text{ \AA}$

Fixed V_c calculations were performed with ESM on1 for both polarities. Apart from a constant, the constraint energy is

$$E_{\text{con}} = \frac{V_c}{2}(N_L - N_R) = -V_c Q. \quad (21)$$

Stationarity at fixed transferred charge gives

$$\frac{\partial E_{\text{DFT}}}{\partial Q} = V_c. \quad (22)$$

The voltage conjugate to charge transfer is therefore evaluated as $V_{\text{total}} = V_c$, and

$$\frac{C_{\text{total}}}{A} = \frac{|Q|e}{|V_c|A}. \quad (23)$$

This is not an instruction to add V_c to the Hartree voltage.

The Hartree geometric capacitance and an effective quantum/electronic capacitance are

$$\frac{C_{\text{geom}}}{A} = \frac{|Q|e}{|\Delta\phi_H|A}, \quad (24)$$

$$\frac{1}{C_Q} = \frac{1}{C_{\text{total}}} - \frac{1}{C_{\text{geom}}}. \quad (25)$$

C_{geom}/A remains constant at approximately $1.0724 \mu\text{F}/\text{cm}^2$, as expected from the linear Poisson response at fixed geometry. C_{total}/A increases with $|V_{\text{total}}|$ and forms a symmetric V-shaped curve. The extracted C_Q is an effective non-Hartree electronic contribution contained in the fixed-potential energy response, rather than a direct density-of-states calculation. The plotted origin follows the paper-style convention

Table 1: Capacitance components in $\mu\text{F}/\text{cm}^2$.

V_c (V)	Q (e)	C_{geom}/A	C_{total}/A	C_Q/A
-0.64	-0.00126618	1.07236	0.59993	1.36177
-0.48	-0.00087786	1.07237	0.55459	1.14860
-0.32	-0.00053004	1.07249	0.50228	0.94472
-0.16	-0.00023935	1.07231	0.45363	0.78624
0.16	0.00023935	1.07231	0.45363	0.78624
0.32	0.00053004	1.07249	0.50228	0.94472
0.48	0.00087786	1.07237	0.55459	1.14860
0.64	0.00126618	1.07237	0.59993	1.36176

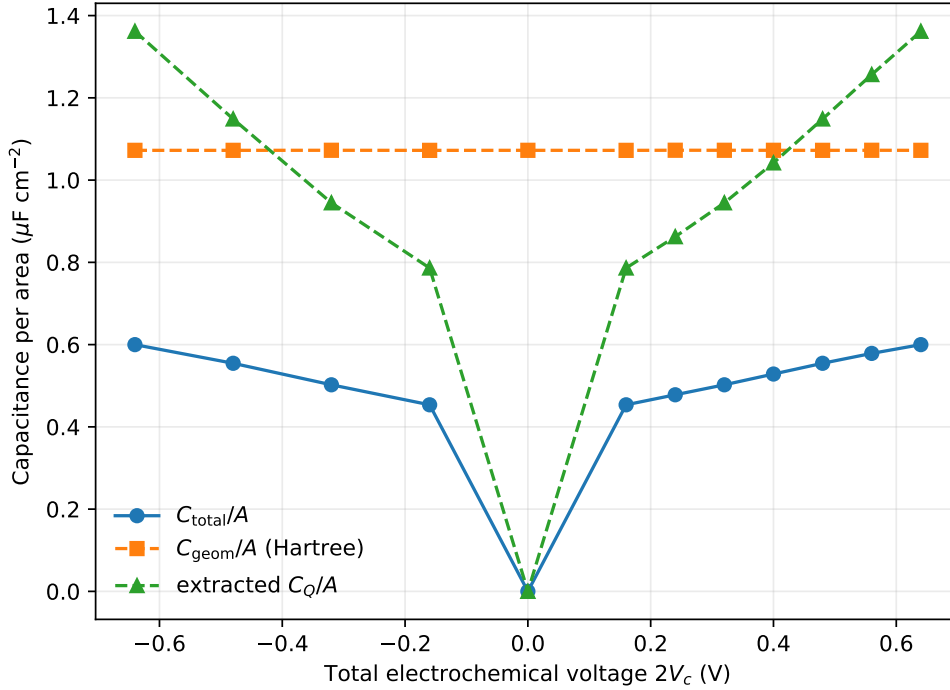


Figure 1: Total, Hartree geometric, and series-extracted quantum/electronic capacitances for $d = 10.05$ Å. The negative branch consists of independent A/B-reversed calculations, not mirrored positive data.

$C_{\text{total}}(0) = 0$; a rigorous zero-bias differential capacitance requires closer-to-zero points and dQ/dV .

For a DOS-based comparison the two electrodes must be treated separately. Let $D_L(E)$ and $D_R(E)$ be layer-projected DOSs per cell in states/eV. Then

$$\frac{C_{Q,L}^{\text{DOS}}}{A} = \frac{e}{A} \int D_L(E) \left[-\frac{\partial f(E - \mu_L, T)}{\partial E} \right] dE, \quad (26)$$

$$\frac{C_{Q,R}^{\text{DOS}}}{A} = \frac{e}{A} \int D_R(E) \left[-\frac{\partial f(E - \mu_R, T)}{\partial E} \right] dE, \quad (27)$$

$$\frac{1}{C_{Q,\text{transfer}}^{\text{DOS}}} = \frac{1}{C_{Q,L}^{\text{DOS}}} + \frac{1}{C_{Q,R}^{\text{DOS}}}, \quad (28)$$

$$C_{Q,\text{transfer}}^{\text{DOS}} = \frac{C_{Q,L}^{\text{DOS}} C_{Q,R}^{\text{DOS}}}{C_{Q,L}^{\text{DOS}} + C_{Q,R}^{\text{DOS}}}. \quad (29)$$

The e/A prefactor applies when energy and DOS use eV; the equivalent SI formula in joules has prefactor e^2/A . For equivalent layers at zero bias, $C_{Q,\text{transfer}}^{\text{DOS}} = C_{Q,\text{layer}}^{\text{DOS}}/2$. Since then $D_{\text{bilayer}} = 2D_{\text{layer}}$,

$$\frac{C_{Q,\text{transfer}}^{\text{DOS}}}{A} = \frac{e}{4A} \int D_{\text{bilayer}}(E) \left[-\frac{\partial f(E - \mu, T)}{\partial E} \right] dE. \quad (30)$$

Using the total bilayer DOS as one electrode would overestimate the transfer capacitance by a factor of four. At short distance the layer DOS is projection-dependent because of hybridization.

We evaluated this comparison at 300 K using zero-bias OpenMX layer PDOSs on a $1 \times 300 \times 300$ k grid and 0.005 eV Gaussian broadening. At every fixed- V_c point, the non-Hartree voltage $V_Q = V_c - \Delta\phi_H$ was mapped symmetrically to $\mu_L = +V_Q/2$ and $\mu_R = -V_Q/2$. The resulting layer capacitances were then combined in series. At $V_c = 0$, the series and DOS results are respectively 3.134 and 3.159 $\mu\text{F}/\text{cm}^2$ for $d = 3.70$ Å, and 0.616 and 0.613 $\mu\text{F}/\text{cm}^2$ for $d = 10.05$ Å. At 0.16 V the corresponding pairs are 3.366 versus 3.164 and 0.953 versus 0.941. Thus the long-distance DOS calculation reproduces both the thermally rounded Dirac minimum and its rise with voltage. At short distance both methods instead give a finite, nearly flat response near 3.1 $\mu\text{F}/\text{cm}^2$, consistent with strong hybridization. Finite-voltage differences include rigid-band, self-consistent band-change, and layer-projection effects.

9 Comparison with the paper and experimental meaning

Table 2 compares the results for $d = 10.05$ Å at a voltage magnitude near 0.6 V. The charge inferred from the published capacitance uses the same primitive-cell area, 5.28353 Å².

Table 2: Comparison with the published result. Capacitances are in $\mu\text{F}/\text{cm}^2$.

Case	C/A	Q (e/cell)
Sohib et al.	~ 2.25	~ 0.00445 (inferred)
ESM-on4 reproduction at 0.584 V	2.7166	0.005233962
ESM-on1 fixed- V_c near 0.6 V	~ 0.59	~ 0.00117
Hartree geometric value	~ 1.0724	~ 0.00117

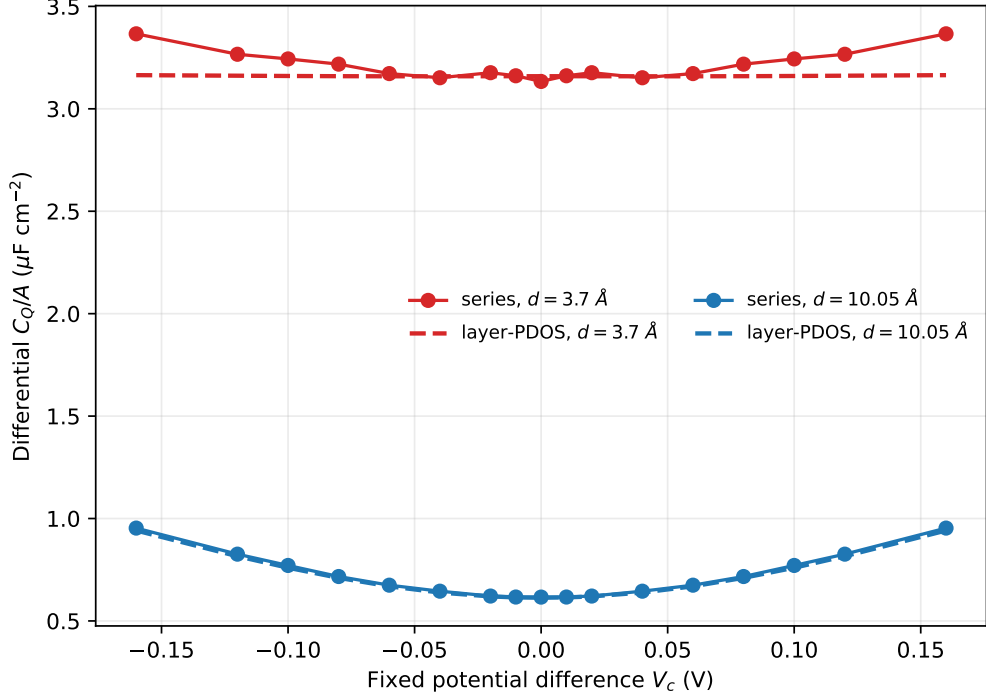


Figure 2: Comparison at 300 K between differential C_Q extracted from the fixed- V_c series relation (solid) and the layer-PDOS transfer capacitance (dashed).

The fixed- V_c total capacitance has the same $1 \mu\text{F}/\text{cm}^2$ order of magnitude and reproduces the symmetric V-shaped trend, but it is only about 26% of the quoted value and is not quantitative agreement. The ESM-on4 reproduction is much closer to the paper. The principal distinction is the boundary condition and voltage definition. ESM on4 with `ESM.potential.diff` imposes an external field across the cell, whereas ESM on1 with the local $+V_c/2, -V_c/2$ potential drives charge transfer directly while removing the nonperiodic image interaction. Furthermore, V_c is the variable conjugate to Q in the fixed-potential functional; it is neither the Hartree potential difference nor a correction to be added to it. Stacking, ESM boundary positions, unreleased author-specific ESM changes, charge partitioning, and the distinction between secant and differential capacitance can add further differences. Carbon SOC is unlikely to account for the approximately fourfold charge discrepancy.

The correspondence to experiment depends on the measurement. The Hartree-only C_{geom} is an electrostatic geometric capacitance. A two-terminal voltage normally measures an electrochemical-potential difference, so its small-signal capacitance contains geometric and quantum contributions in series:

$$C_{\text{meas}} = \frac{dQ}{dV_{\text{electrochemical}}}, \quad \frac{1}{C_{\text{meas}}} = \frac{1}{C_{\text{geom}}} + \frac{1}{C_Q}. \quad (31)$$

In this sense, the ESM-on1 constrained calculation is conceptually closer to an image-free two-terminal capacitor, whereas ESM on4 is the appropriate comparison for the paper’s externally driven protocol. A direct small-signal comparison must use $dQ/d(V_c)$ rather than the finite-bias secant $Q/(V_c)$ and must test the Mulliken group dependence. Neither the present C_{total} nor the paper’s Hartree-based ratio should therefore be identified with an experimental terminal capacitance without these qualifications.

10 Stock-OpenMX reanalysis of the Sohib protocol

We performed a stock OpenMX 3.9.9 ESM-on4 sweep at 300 K for six distances $d = 3.70, 5.82, 7.93, 10.05, 12.17,$ and $14.29 \text{ \AA}$ and for $V_{\text{ESM}} = 0.4, 0.8, 1.2, 1.6, 2.0,$ and 2.6 V , using the paper's $1 \times 100 \times 100$ SCF k grid. Following Sohib et al., the operational total capacitance is

$$\frac{C_{\text{total}}^{\text{Sohib}}}{A} = \frac{|Q|e}{A|\Delta V_H^{\text{raw}}|}, \quad (32)$$

where ΔV_H^{raw} is the layer-to-layer difference read directly from the OpenMX .vhart cube. The layer-PDOS capacitances at 300 K are

$$C_{Q,i}^{\text{DOS}} = \frac{e}{\int D_i(F)} \left[-\frac{\partial f(E - \mu_i, T)}{\partial F_i} \right] dF_i. \quad (33)$$

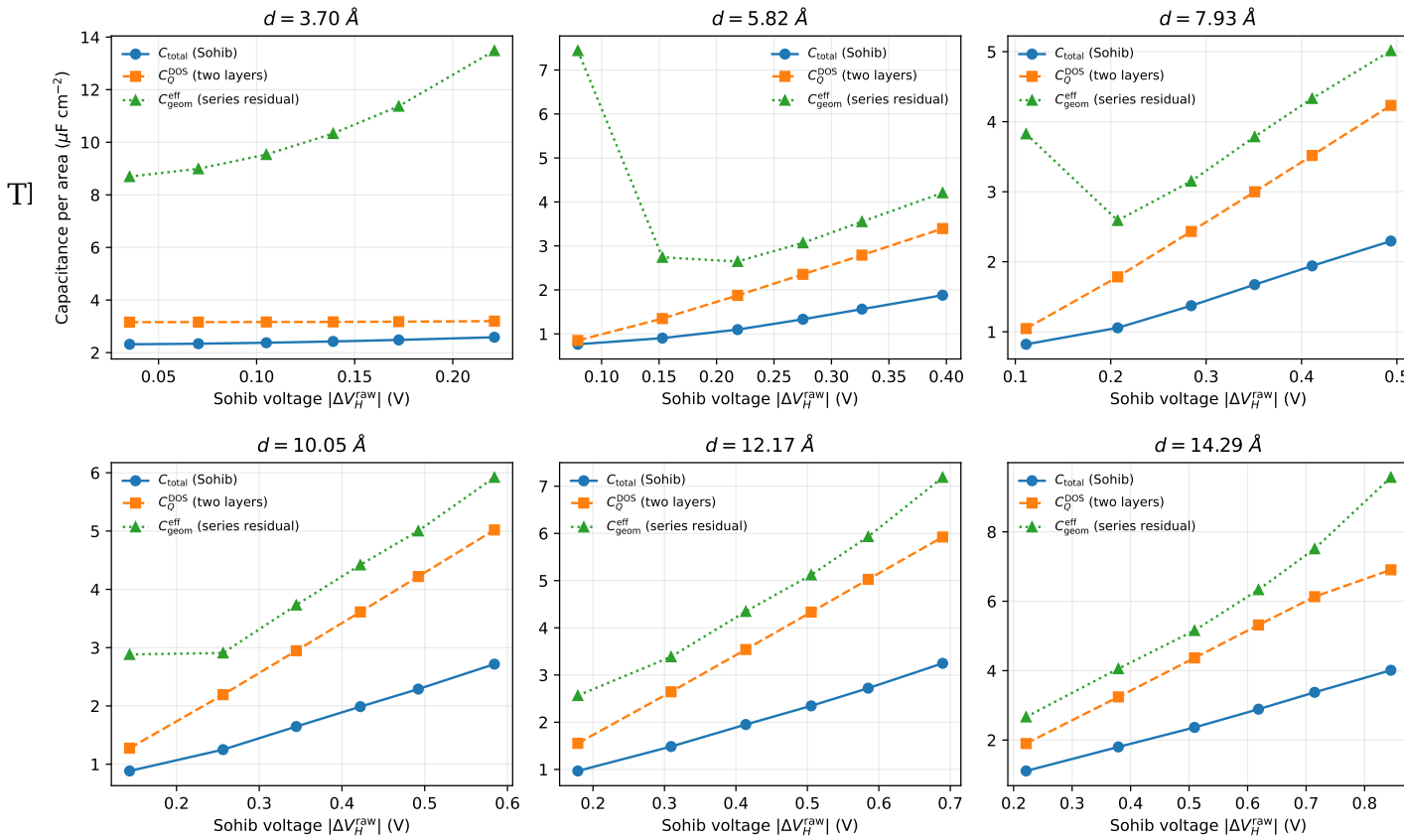


Figure 3: Stock-OpenMX ESM-on4 reanalysis. The total follows the raw-Hartree definition of Sohib et al.; C_Q is obtained from the two layer DOSs, and $C_{\text{geom}}^{\text{eff}}$ is the algebraic series residual.

At $V_{\text{ESM}} = 2.6 \text{ V}$, the triples $(C_{\text{total}}^{\text{Sohib}}, C_{Q,\text{transfer}}^{\text{DOS}}, C_{\text{geom}}^{\text{eff}})/A$ for increasing d are $(2.584, 3.197, 13.484), (1.879, 3.395, 4.209), (2.295, 4.232, 5.014), (2.717, 5.019, 5.922), (3.247, 5.923, 7.188),$ and $(4.011, 6.906, 9.571)$, in $\mu\text{F}/\text{cm}^2$. These residual values are not classical vacuum geometric capacitances. Stock OpenMX adds the external ESM-on4 ramp directly to dVHart_Grid_B, so the raw cube contains both external and induced terms. After subtracting the analytic external ramp, the induced-Hartree diagnostic gives 2.201, 2.072, 2.109, 2.197, 2.507, and $3.347 \mu\text{F}/\text{cm}^2$. The signed external difference is $\Delta V_{\text{ext}}^{R-L} = -V_{\text{ESM}}d/L$, as follows from the $-0.5V_{\text{ESM}}(z - z_1)/z_1$ term in Poisson_ESM.c; ϵ_0/d gives 2.393, 1.521, 1.116, 0.881, 0.728, and $0.620 \mu\text{F}/\text{cm}^2$.

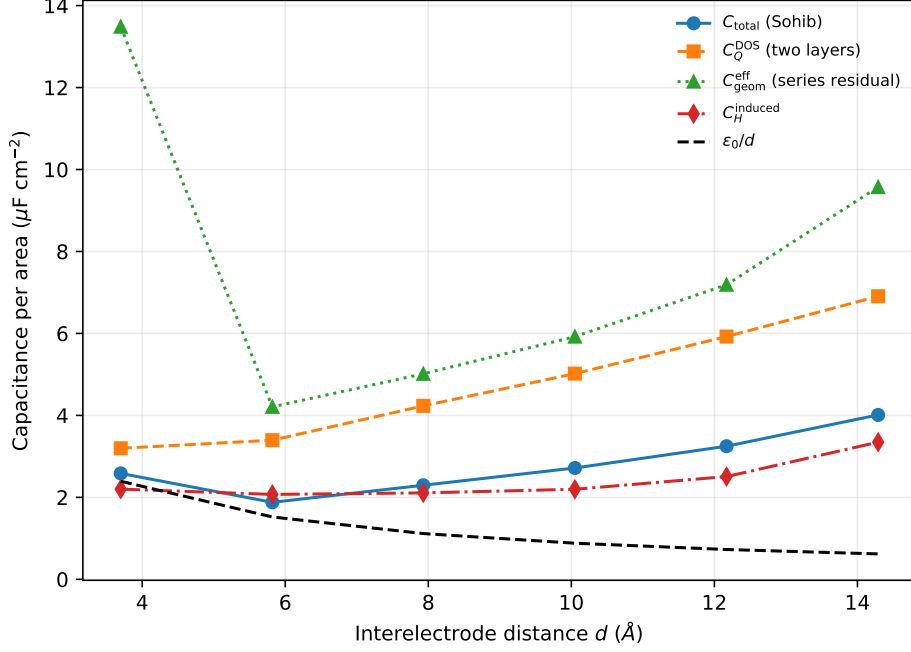


Figure 4: Distance dependence at $V_{\text{ESM}} = 2.6$ V.

This diagnostic is not a pure geometric capacitance across the series: short-distance overlap and charge partitioning matter, while at $d = 14.29$ Å the sheet-ESM clearance is only 2.855 Å in the 20 Å cell and boundary/contact effects contaminate the endpoint. Thus the agreement near 10.05 Å is not a general validation, and $C_{\text{geom}}^{\text{eff}}$ is more accurately an effective other/series capacitance.

10.1 ESM-boundary convergence

We repeated the three largest distances in a 30 Å cell. Because fixed V_{ESM} would reduce the applied field when increasing the cell, the 30 Å values were scaled by $30/20 = 1.5$; hence 2.6 V in the 20 Å cell is compared with 3.9 V in the 30 Å cell. This keeps both the external field and $V_{\text{ESM}}d/L$ fixed.

d (Å)	edge ₂₀	edge ₃₀	$C_{H,20}^{\text{ind}}/A$	$C_{H,30}^{\text{ind}}/A$	ϵ_0/d
10.05	4.975	9.975	2.197	1.420	0.881
12.17	3.915	8.915	2.507	1.357	0.728
14.29	2.855	7.855	3.347	1.346	0.620

Capacitances are in $\mu\text{F}/\text{cm}^2$. Increasing the clearance reduces the 14.29 Å value from 3.347 to 1.346 , confirming strong ESM-boundary contamination in the 20 Å cell. The nearly flat 30 Å trend remains about 1.5 – 2.2 times ϵ_0/d , so this Hartree diagnostic is not the ideal parallel-plate capacitance.

11 Two distances: analogues of Fig. 4(b) and 4(c)

The same ESM-on1 fixed- V_c sweep was performed for $d = 3.70$ and 10.05 Å. At $|V_c| = 0.6$ V, linear interpolation gives

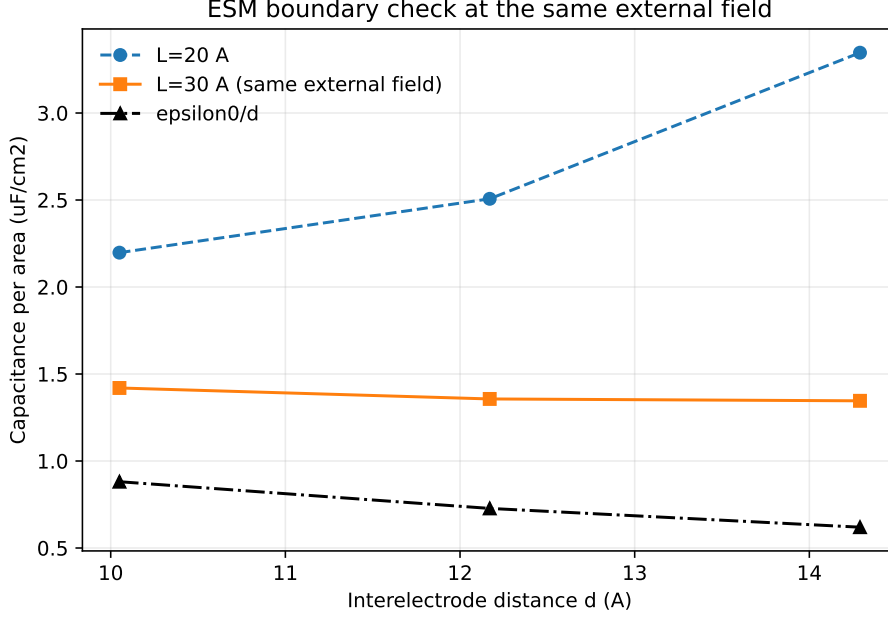


Figure 5: ESM-boundary convergence at the same external field.

d (Å)	Q (e/cell)	C_{total}/A	C_{geom}/A	C_Q/A
3.70	0.00469	2.37	5.98	3.92
10.05	0.00117	0.589	1.072	1.31

Capacitances are in $\mu\text{F}/\text{cm}^2$. The paper reports approximately 3.4 and 2.25, respectively, so the present totals are about 70% and 26% of those values. The short-distance C_{geom} decreases mildly from 6.22 to 5.96 over the sweep, indicating interlayer-overlap effects; it remains nearly constant at 1.0724 for 10.05 Å.

As in the paper, the figures separate C_{total} and C_Q . The A/B-symmetric negative branch and the display origin make each V shape explicit. C_{geom} is retained in the separate diagnostic file `graphene_two_distance_cgeom.pdf`.

The extracted C_Q has a V-like shape resembling Fig. 4(c), but it is not the same quantity. The paper thermally broadens a high-k-density DOS, whereas the present value is a residual defined by the series equation. It contains non-Hartree fixed-potential response and, especially at 3.70 Å, interlayer hybridization and charge-partition effects. Its strong distance dependence demonstrates the distinction. The plotted origin is the paper-style convention; because the closest computed point is $|V_c| = 0.16$ V, the zero-bias value and differential slope have not been determined.

12 Small- V_c validation of the Dirac-band response

The earlier plot assigned zero capacitance at the origin only as a display convention. We therefore calculated $V_c = 0.01, 0.02, 0.04, 0.06, 0.08, 0.10, 0.12$, and 0.16 V for both distances. A $1 \times 300 \times 300$ k grid, chosen as a multiple of three to resolve the K region, a 300 K electronic temperature, and a 10^{-8} Hartree SCF criterion were used. The negative branch follows from A/B exchange symmetry. The symmetric

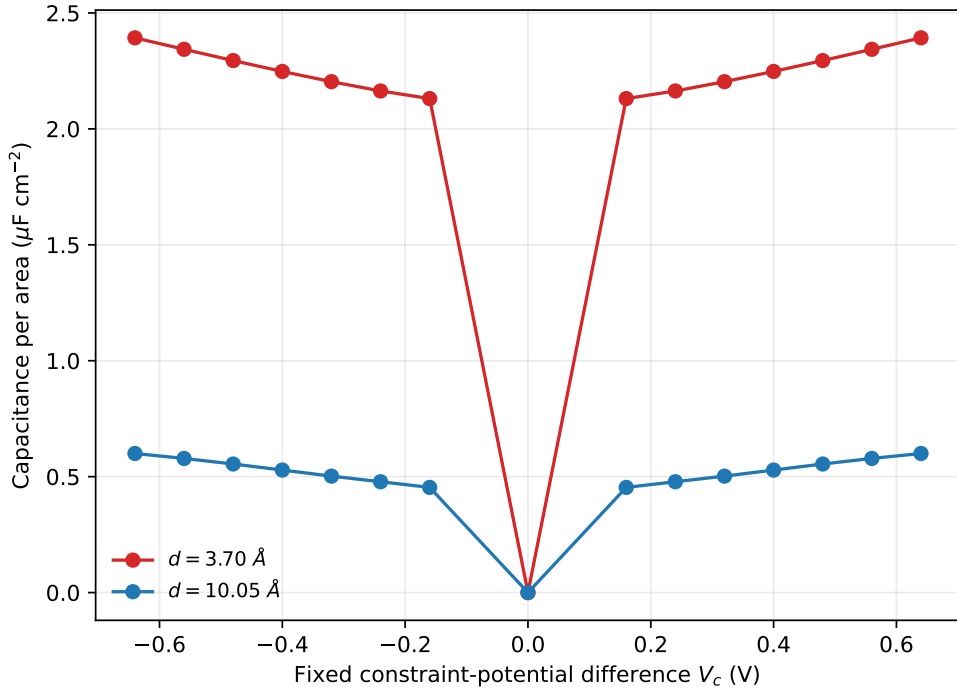


Figure 6: Fig. 4(b)-like C_{total} . The geometric capacitance is omitted so that the panel contains only the paper-comparable total quantity.

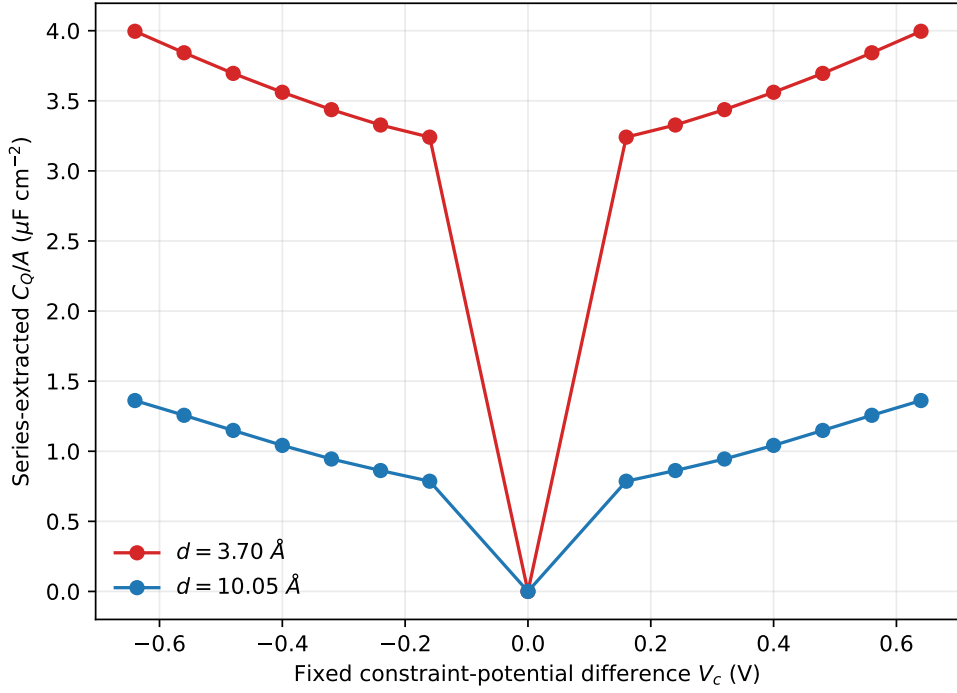


Figure 7: Effective C_Q extracted from the series relation.

data were differentiated as

$$\frac{C_{\text{total}}^{\text{diff}}}{A} = \frac{e}{A} \frac{dQ}{dV_c}, \quad (37)$$

$$\frac{C_H^{\text{diff}}}{A} = \frac{e}{A} \frac{dQ}{d\Delta\phi_H}, \quad (38)$$

$$\frac{1}{C_Q^{\text{diff}}} = \frac{1}{C_{\text{total}}^{\text{diff}}} - \frac{1}{C_H^{\text{diff}}}. \quad (39)$$

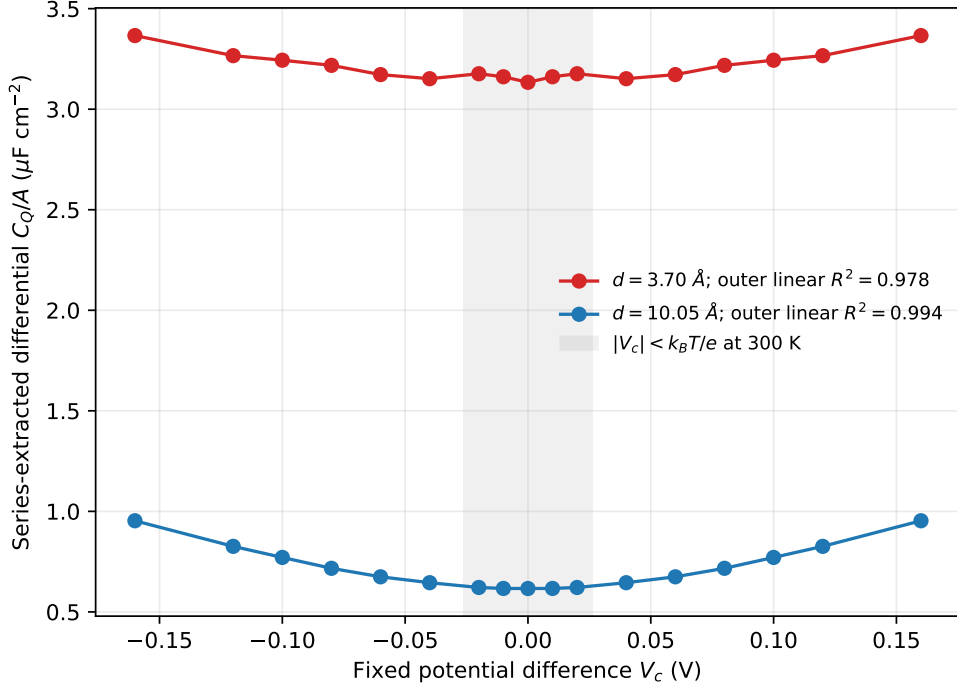


Figure 8: Differential effective C_Q from the dense small- V_c sweep. The shaded interval is $|V_c| < k_B T / e = 0.026 \text{ V}$ at 300 K.

For $d = 10.05 \text{ \AA}$, C_Q^{diff}/A is $0.616 \mu\text{F/cm}^2$ at zero and $0.953 \mu\text{F/cm}^2$ at 0.16 V . A linear fit over $0.06\text{--}0.16 \text{ V}$ gives $R^2 = 0.994$. The finite rounded minimum on the thermal scale followed by an almost linear rise with $|V_c|$ is the expected thermally broadened signature of a graphene Dirac DOS. For $d = 3.70 \text{ \AA}$, the corresponding values are 3.134 and 3.366 , and the response is comparatively flat; interlayer overlap and hybridization suppress the isolated-Dirac signature. The fixed-potential series extraction therefore detects the Dirac character in the weakly coupled system and its loss at short separation, although it is not a direct DOS integration.

13 Zero-temperature estimate from the series relation

The calculation was repeated at an electronic temperature of exactly 0 K , using ESM on1, the same $1 \times 300 \times 300 \text{ k}$ grid and a 10^{-8} Hartree SCF criterion. The sampled voltages were $V_c = 0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.12, \text{ and } 0.16 \text{ V}$. No DOS expression is used in the following estimate. For each finite-voltage calculation we

define the secant quantities

$$\frac{C_{\text{total}}}{A} = \frac{|Q|e}{A|V_c|}, \quad \frac{C_H}{A} = \frac{|Q|e}{A|\Delta\phi_H|}, \quad (40)$$

$$\frac{1}{C_Q} = \frac{1}{C_{\text{total}}} - \frac{1}{C_H}. \quad (41)$$

Hence the value actually evaluated by the script is

$$\frac{C_Q}{A} = \frac{|Q|e}{A(|V_c| - |\Delta\phi_H|)}. \quad (42)$$

C_H is the effective Hartree electrostatic capacitance, while C_Q is the effective non-Hartree residual defined by this series decomposition. These secant capacitances are distinct from the differential quantities of the preceding section.

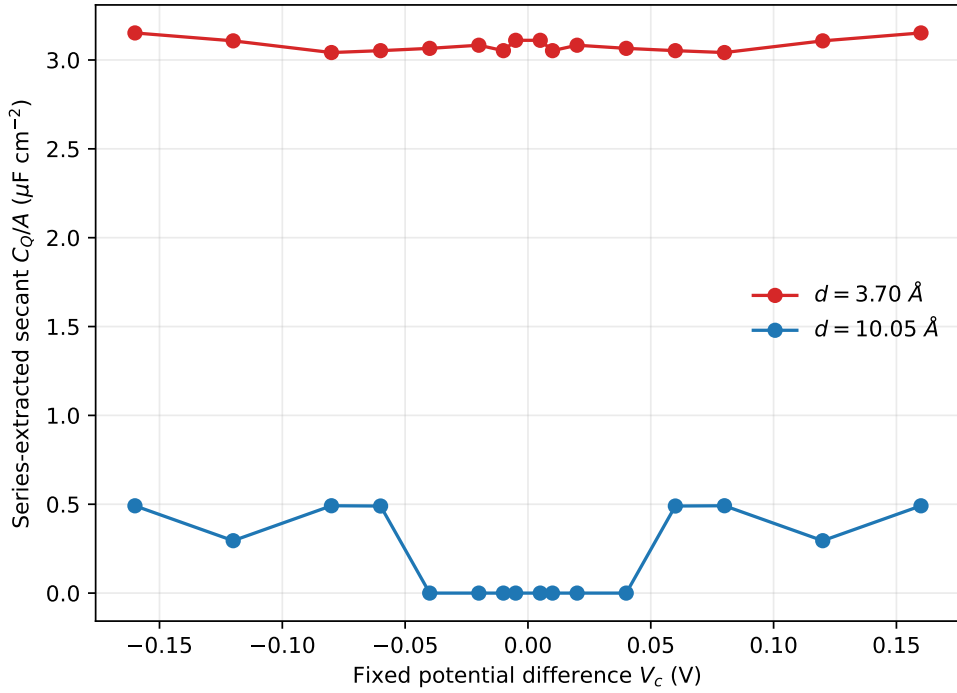


Figure 9: Zero-temperature secant C_Q obtained solely from the series relation. The negative branch is generated by A/B exchange symmetry.

For $d = 3.70 \text{ \AA}$, C_Q/A lies between 3.04 and $3.15 \mu\text{F/cm}^2$ and is nearly independent of voltage, as expected when interlayer overlap and hybridization are strong. The long-distance results are

V_c (V)	Q (e/cell)	C_Q/A ($\mu\text{F/cm}^2$)
0.005-0.04	0	0
0.06	0.00008889	0.490
0.08	0.00008889	0.492
0.12	0.00008889	0.295
0.16	0.00017778	0.491

The staircase at $d = 10.05 \text{ \AA}$ is a finite-k sampling effect rather than a physical gap. At exactly zero temperature, occupations on a finite mesh change only when a discrete sampled state crosses the Fermi level. Thus the continuum Dirac limit $C_Q \rightarrow 0$

as $V_c \rightarrow 0$ appears as a zero-charge interval followed by discrete steps. A smooth $T=0$ V-shaped differential capacitance requires systematic k-grid convergence; differentiating the present staircase would not be meaningful.

14 Validation and limitations

For a two-layer graphene test with ESM on1 and $d = 3.70$ Å, the calculation gave $Q = 0.00188598$, $|\Delta\phi_H| = 0.09225$ V, and $C/A = 6.199$ $\mu\text{F}/\text{cm}^2$.

The current limitations are:

1. no constraint contribution to forces, stress, or total energy;
2. no persistence of reference populations and V_c in restart files;
3. basis dependence of Mulliken populations;
4. no NEGF or multiple independent constraints;
5. a need to distinguish Hartree, external constraint, and electrochemical voltage definitions.